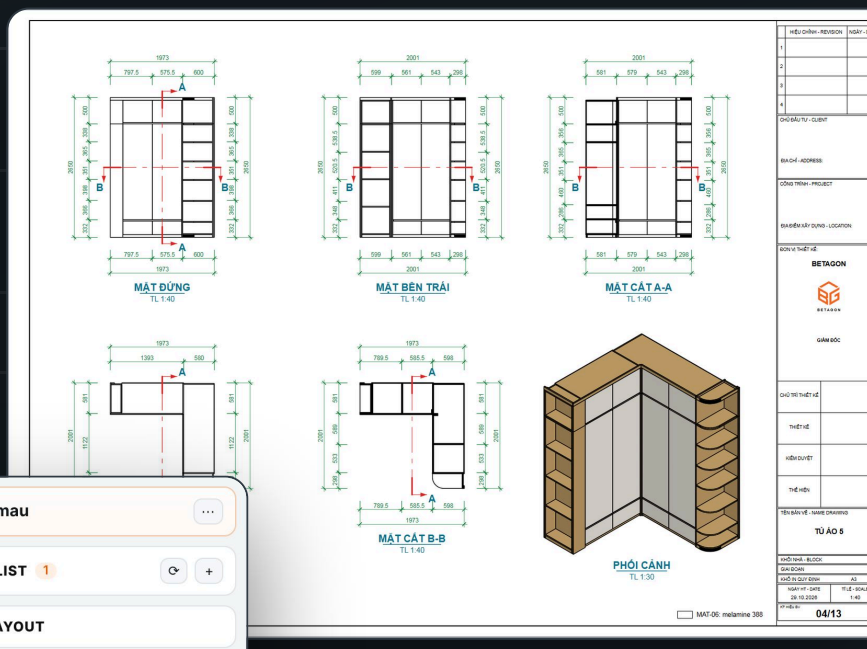
BETAGON · SKETCHUP PLUGIN

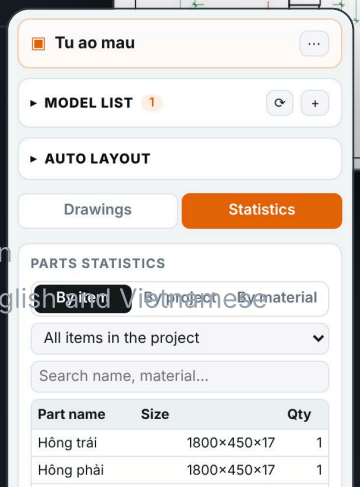
BETAGON LAYOUT

Complete user guide

Version 1.3.15 **3-DAY FREE TRIAL**

- Automatic furniture drawing sheets from your SketchUp model
- Front, side, plan, section and 3D views, enlarged details
- Automatic dims, notes, 2D symbols, material hatches, title block
- Parts statistics with Excel export
- DXF that opens straight in AutoCAD, true-scale PDF





Author: **Nguyễn Văn Thanh** · Betagon

Supports SketchUp 2021 – 2026 · English and Vietnamese

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1 Introduction

Betagon Layout lays out furniture drawings right inside SketchUp. Select cabinets, shelves, tables... in your model and click **Add**: the plugin builds the front, side, plan, section and 3D views, places the dimensions and the title block on the sheet. You then fine-tune by hand, much like in LayOut, and export DXF for AutoCAD or PDF for printing.

Automatic layout

Each item gets its own set of views, a scale that fits the paper, and a second sheet when needed.

Dims and notes

Automatic detail and overall dims; linear, aligned, angle and row dims; leader notes, door/compartment notes.

Sections and details

Vertical and horizontal sections with hatch; drag the section marker to move the cut; circle an area to extract an enlarged detail.

Parts statistics

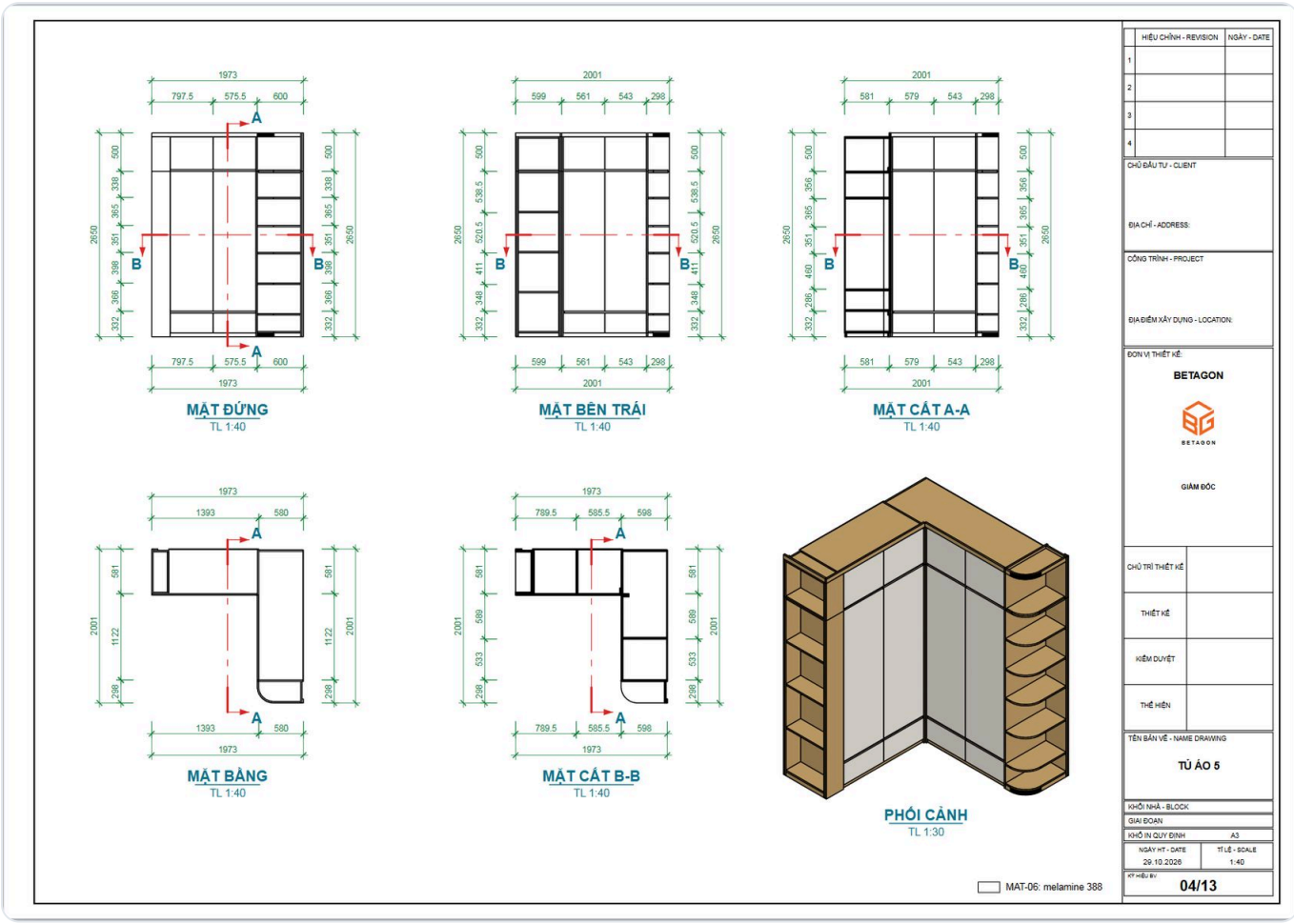
Part lists by item, by project and by material, with area and volume; Excel export.

DXF and PDF export

AutoCAD 2010 DXF at true 1:1 scale in mm; sharp vector PDF that prints to scale.

English and Vietnamese

Switch language in Settings. Runs on SketchUp 2021 to 2026.



Shop drawing of an L-shaped corner wardrobe laid out automatically in a real project: front, left side, section A-A, plan, section B-B, material-colour perspective and the material legend in the sheet corner

2 Installing and updating

Requirements

- SketchUp 2021, 2022, 2023, 2024, 2025 or 2026 on Windows.
- Installer file: **Betagon_Layout_1.3.15.rbz**.
- **Betagon Center** (free) to sign in, start the trial, buy a license and get updates.
- An internet connection when Layout opens, to check the license.

Option 1: install from Betagon Center (recommended)

1. Open **Betagon Center** in SketchUp and sign in to your Betagon account.
2. Find **Betagon Layout** and click **Install** (or **Update** if an older version is installed).
3. The **Betagon Layout** toolbar appears with its buttons.

Option 2: install the .rbz with Extension Manager

1. Go to **Extensions › Extension Manager** and click **Install Extension**.
2. Pick the **.rbz** file and click **Yes** to confirm.
3. If the toolbar does not show, go to **View › Toolbars** and tick **Betagon Layout**.

> **Installing a new version**
>
> Install the new version over the running one: Layout reloads its code after a few seconds, no SketchUp restart needed. If the window reports a version mismatch, use **Extensions › Betagon Layout › Reload plugin**; if it still fails, close SketchUp completely and open it again.

3 Trial and license

Plan	Price	Notes
Trial	Free	3 days with every feature, once per computer.
1 computer	129,000 VND	Perpetual license, activated on 1 computer.
2 computers	199,000 VND	Perpetual license, activated on 2 computers.

1. The first time you open Layout on an account that has not used the trial, the plugin offers the **free 3-day trial**. Click **Yes** and start working.
2. The remaining time shows in **Settings · Info › License**.
3. When the trial ends, Layout saves your drawing, closes the window and opens the purchase page in Betagon Center. You can click **Buy license** in Settings at any time.
4. After paying, click **Check again** in Settings (or reopen Layout) to receive the license.

> **Not signed in**
>
> Layout needs a Betagon account signed in to Betagon Center. If it asks you to sign in, open Center, sign in, then open Layout again.

4 SketchUp toolbar buttons

Button	Name	What it does
	**Drawing layout editor**	Opens the Betagon Layout window. It is a separate application window: you can keep it on a second screen and Alt+Tab between it and SketchUp.
	**Quick DXF export**	Select furniture items (Groups or Components) and click: the plugin lays them out automatically and saves one DXF file straight away, without opening the Layout window.
	**User guide**	Opens this guide (a PDF shipped with the plugin) in the current language: English or Vietnamese.
	**Settings · Info**	Opens the settings window: language, license, auto save, shortcuts, data folder, Zalo support group.

These commands are also in the **Extensions › Betagon Layout** menu, together with **Reload plugin (after installing a new version)**.

5 Preparing the model

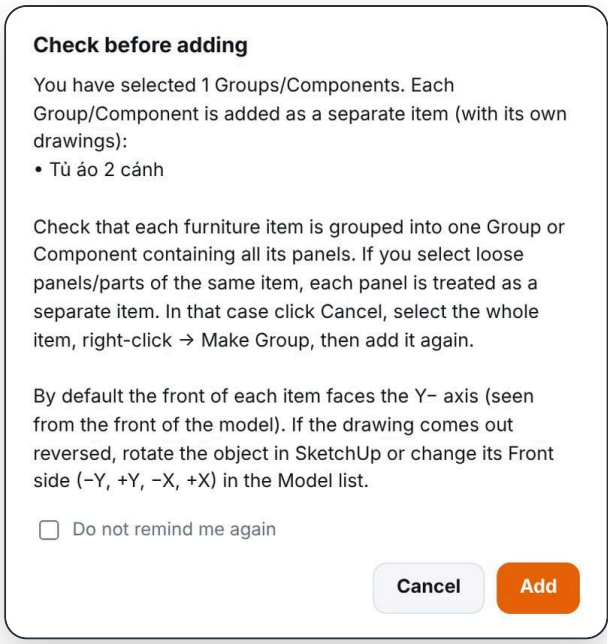
Layout treats **each selected Group or Component as one furniture item**. Inside the item, each child Group/Component is a **part** (side panel, top, bottom, door...).

- Wrap the whole cabinet in **1 Group or Component** containing all of its panels. Name the Group (for example "Wardrobe 2 doors"): that name becomes the item name and the drawing name.
- Each panel should be a named child Group/Component ("Left side", "Right door"...) with a material: the statistics take names and materials from here.
- If you select loose panels of one cabinet, each panel is treated as a separate item. Loose faces and edges outside any Group/Component are ignored.

Each time you click **Add**, Layout shows **Check before adding**: it lists the Groups/Components that will each become an item and reminds you to group loose panels. If the selection is wrong, click **Cancel**, select the whole item, right-click › **Make Group** and add again.

Once you are used to it, tick **Do not remind me again**. The box still appears when the selection contains loose faces or edges.

The item's **front side** defaults to the Group's −Y side (seen from the front of the model). The reminder says so too: if a drawing comes out reversed, rotate the object in SketchUp or change the item's **Front side** in the Model list.



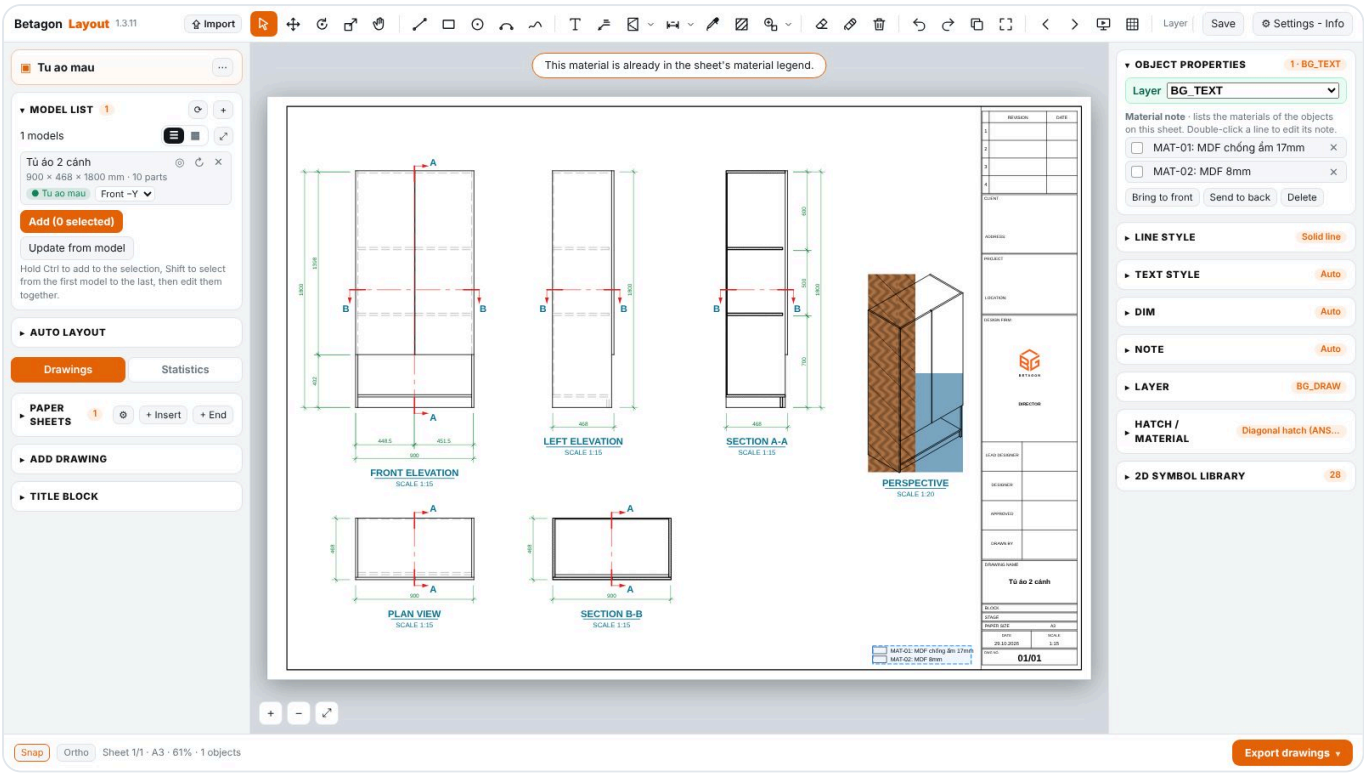
Reminder before adding

6 Quick start

1. Click **Drawing layout editor** on the toolbar. The first time, Layout creates a project named after the SketchUp file.
2. In SketchUp, select one or more furniture items (Groups/Components).
3. Click **Add (n selected)** in the **Model list** panel or the **+ n** button on the panel header, check the reminder, then click **Add**.
4. Layout builds the views, places dims, picks the scale and arranges everything on the sheet. Each item gets at least one sheet.
5. Fine-tune if needed: move viewports, add dims, notes, symbols, hatches, edit the title block.
6. Click **Export drawings ▾** at the bottom right and choose **PDF**, **CAD (DXF)** or **Image**.

> **Editing the model after layout**
>
> After changing a cabinet in SketchUp, click **Update from model** (or the ⟳ button). Layout only redraws the views and dims of items that changed; dims and notes you added yourself are kept.

7 The workspace



The Betagon Layout window: model list and sheets on the left, the open sheet in the middle, properties and style sets on the right

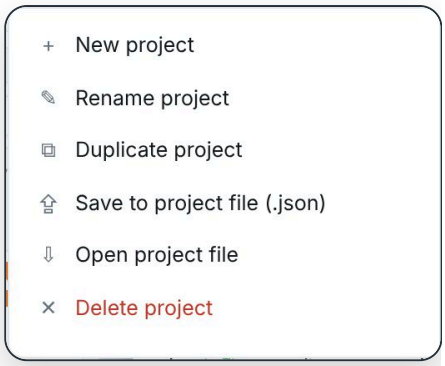
Area	Contents
Top bar	Import, drawing and dimension tools, undo/redo, sheet navigation, presentation, grid, layer for new objects, **Save** and **Settings - Info**.
Left column	Project, Model list, Auto layout, the **Drawings** tab (Paper sheets, Add viewport, Title block) and the **Statistics** tab.
Center	The open sheet. Scroll to zoom, hold the middle button to pan, + − ⤢ buttons in the bottom corner.
Right column	Object properties, style sets (Line style, Text style, Dim, Note), Layers, Hatch / material, 2D symbol library.
Bottom bar	**Snap** and **Ortho** toggles, cursor coordinates, the **Export drawings ▾** button (PDF, CAD, Image).

Hover over a setting to see a short explanation. You can turn these popups off in Settings.

8 Projects

Each **project** is a separate file on your computer holding the model list, drawing sheets and style sets. One project can contain items from several SketchUp files. The project name is at the top of the left column; click ⋯ for the menu:

- **New project**: an empty project that keeps the company, drafter and style sets of the current one.
- **Rename project**, **Duplicate project**.
- **Save to project file**: exports the whole project (drawings and built views) to a **.bglayout** file to open on another computer, even one without the original SketchUp file.
- **Open project file**: imports a .bglayout file from another computer.
- **Delete project**: removes it from this computer; the SketchUp file is not affected.
- **Switch project**: the list of projects on this computer.



The project ⋯ menu

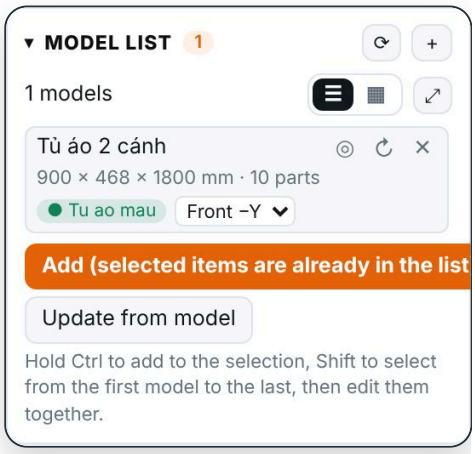
Saving and automatic backups

Auto Save is on by default: every change is saved to the project at once, and every 10 minutes (choose 5, 10, 15 or 30) Layout also makes an automatic backup, keeping the latest 12. Use **Settings › Restore a backup...** to go back to an older one; the current state is saved before it is replaced. With Auto Save off, the project is only saved when you press Ctrl + S or click **Save** (the top bar shows "● Not saved").

> **Where is my data?**
> Projects are stored in **%APPDATA%\Betagon\Layout** (open it quickly with **Open data folder** in Settings). Built views are saved with the project, so items from another SketchUp file still display correctly when that file is not open.

9 Model list

Each row is an added item: name, bounding size, part count, the SketchUp file it belongs to (green dot = file currently open) and its front side.



Model list

- **Click the name** to rename the item. A name you type is kept and is not overwritten by the SketchUp name.
- ◎ selects and zooms to the item in SketchUp.
- ↻ **Re-layout**: rebuilds the item's viewports and automatic dims from scratch.
- × removes the item from the project (with its viewports and dims).
- **Front side**: Front −Y, +Y, −X, +X. Change it when a cabinet was modelled facing backwards or sideways.
- **Update from model** (⟳): re-reads the geometry after you edit the model.

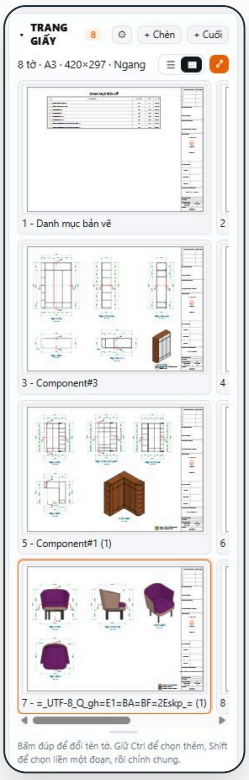
Hold Ctrl to add to the selection or Shift for a range, then Re-layout, Select in SketchUp or Remove from project for the whole group.

Thumbnail view

The ☰ / ▦ buttons switch between a list and **thumbnails**. Each model's thumbnail is the perspective view from the drawing, with material colours. When the other panels are closed, the list grows to the full height of the left column; drag the grip at the bottom to resize it.

The **+** button in the header **turns orange** when SketchUp has models selected that are not in the list yet, and counts only those new models.

Update (⟳) rebuilds only the objects you edited or turned in SketchUp; unchanged objects are kept, so it is fast. Very heavy objects (many curved faces, meshes) are offered **Light display**: outline only, much faster.

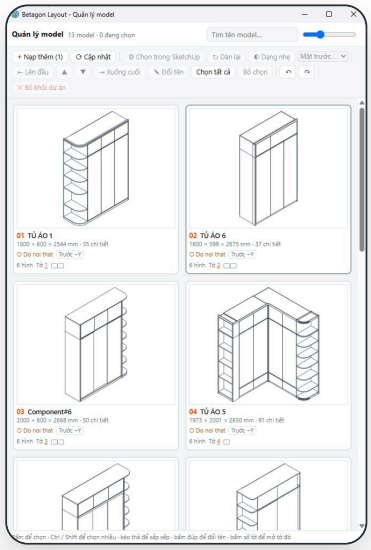


Sheets as thumbnails (Vietnamese UI)

Model manager window

The ⤢ button opens the model list in its own resizable window; the slider at the top sets the thumbnail size.

- Click to select, `Ctrl` / `Shift` for several, **drag and drop** to reorder, double-click to rename.
- Command bar: **Add**, **Update**, **Select in SketchUp**, **Re-layout**, **Light display**, **Front** direction, move to top / bottom, undo, **Remove from project**.
- Each card shows the overall size, part count, SketchUp file, number of views and **sheet numbers**: click a sheet number to open it in Layout.



Model manager window (⤢ button, Vietnamese UI)

10 Auto layout

These settings are used when adding new items, on Re-layout and on **Apply to all drawings**.

Drawings

Tick the views you need: plan, front, left side, right side, back, section A-A (vertical), section B-B (horizontal), 3D view. Each view has a **Display style**: **Black and white** or **Material colors** (colors and

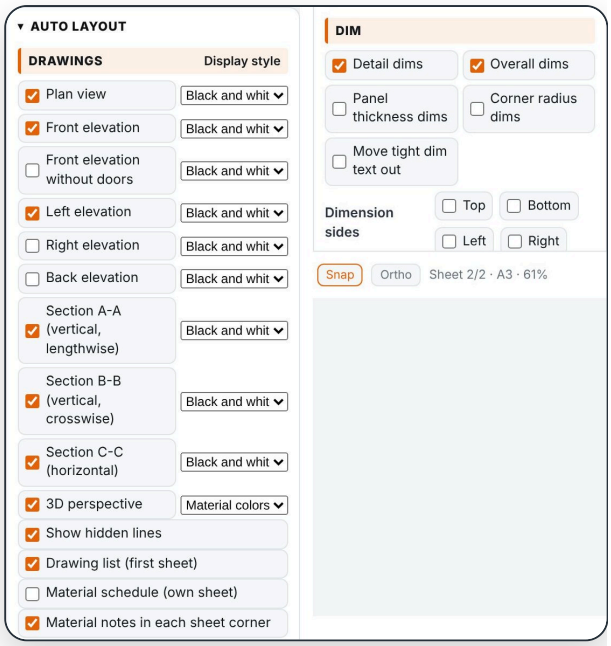
textures from SketchUp). **Show hidden lines** draws dashed lines for hidden parts.

Dims

Detail dims measure every part segment, **Overall dims** give the outer size. **Dimension sides**: tick Top, Bottom, Left, Right; with none ticked, Layout chooses (bottom and one side).

Sheets and scale

- **Number of sheets**: Automatic, Combine on 1 sheet, Split into 2 sheets (separate sections), 1 sheet per drawing.
- **Scale**: Automatic = the largest scale that fits the paper, or a fixed 1:10, 1:15, 1:20...



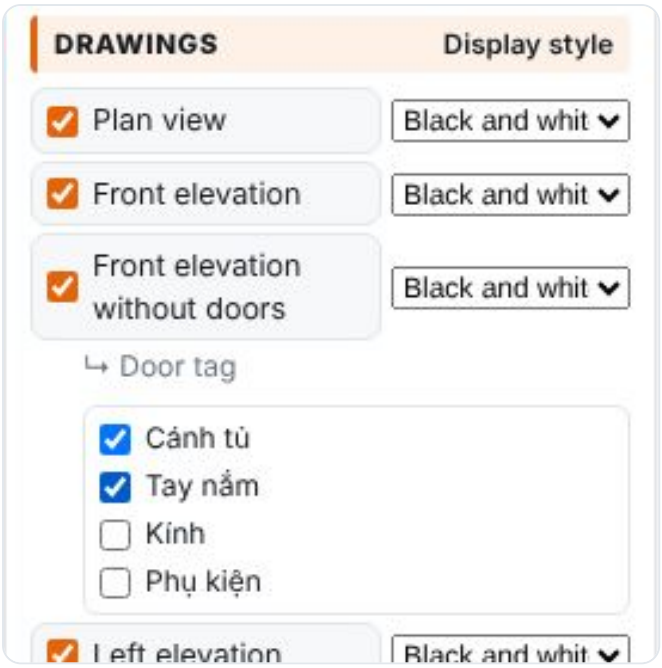
The Auto layout panel

Apply to all drawings

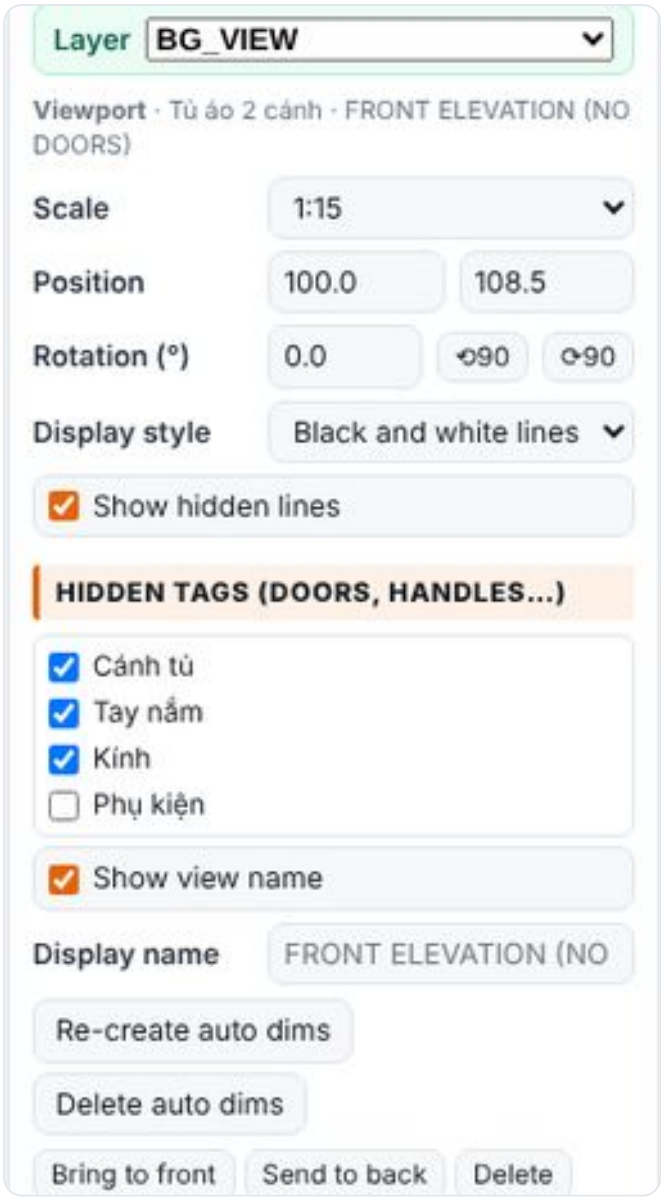
Re-lays out every item of the open file from scratch with the new settings. Tick **Keep hand-added dims and notes** to keep what you drew yourself (re-attached to the right view); untick to clear it. Press Ctrl + Z to undo.

More options since 1.3

- **Sections**: A-A vertical lengthwise, B-B vertical crosswise, C-C horizontal (plan section). Drag a section marker on the drawing to move the cut.
- **Elevation without doors**: an extra front view with doors and drawers hidden to show the carcass. Put doors, handles, glass... on SketchUp tags and **tick one or more tags** to hide (pictures below).
- Automatic **radius dims** for rounded corners and arcs: one dim per corner, the number sits beside the shape on a curved, draggable leader.
- **Dim number jump** (off by default): numbers that do not fit move outside.
- **Material schedule** (own sheet) and the **material legend** in each sheet corner (on by default), see section 18.



Auto layout: tick several tags to hide in the doorless elevation



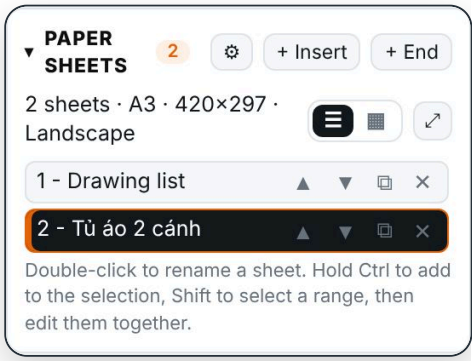
Select a doorless elevation: tick or untick tags in its properties, the view redraws at once

The tag list also appears in **Add drawing** when the view is *Front elevation without doors*. Right-click an elevation › **Hide doors** uses the tags ticked in Auto layout.

11 Paper sheets

The **Drawings › Paper sheets** tab lists the sheets. Click a sheet to open it; double-click to rename; ▲ ▼ reorder; ⧉ duplicate; × delete. **+ Insert** adds a sheet after the selected one, **+ End** adds one at the end. The ☰ / ▦ button switches between list and thumbnails.

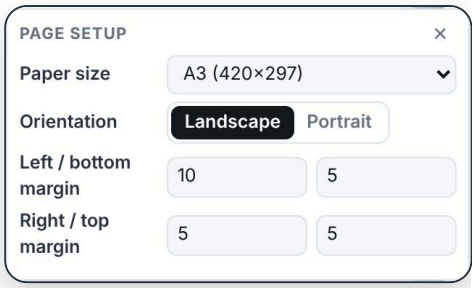
Hold `Ctrl` or `Shift` to select several sheets, then Export CAD, Export PDF, Duplicate or Delete them together.



Sheet list

The ⚙ button opens **Page setup**: paper size A4 to A0 or a custom width × height, Landscape / Portrait orientation and frame margins (left, bottom, right, top, in mm). The paper size applies to every sheet in the project.

After changing the paper size, click ↻ on an item in the list to rearrange its views to fit.



Page setup (⚙ button)

Moving around the sheet

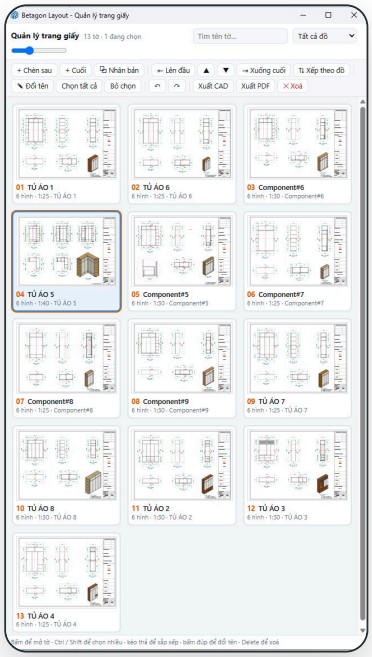
- Scroll wheel: zoom around the cursor. Hold the middle button or use the **Pan** tool (`H`) to pan.
- `F` or the ⤢ button: fit the sheet. `PageUp` / `PageDown`: previous / next sheet.
- `F5` **Presentation**: the sheet full screen, left/right arrows change sheets, `Esc` exits.
- `Ctrl` + `'` **Grid**: a faint grid to align drawings; it is never printed or exported.

Sheet manager window

The ⤢ button on the Paper sheets panel opens all sheets in their own window, with a name search and a filter by object.

- Click to open a sheet, `Ctrl` / `Shift` for several, drag and drop to reorder, double-click to rename, `Delete` to delete.
- Command bar: **+ Insert after**, **+ End**, **Duplicate**, move to top / bottom, **Sort by object**, **Rename**, **Export CAD**, **Export PDF**, **Delete**.
- Each card shows the number of views, the scale and the object of the sheet. The window stays above Layout and can sit on a second screen.

Sheet names you changed are kept when you re-lay out an object.


Sheet manager window (⤢ button, Vietnamese UI)

12 Title block

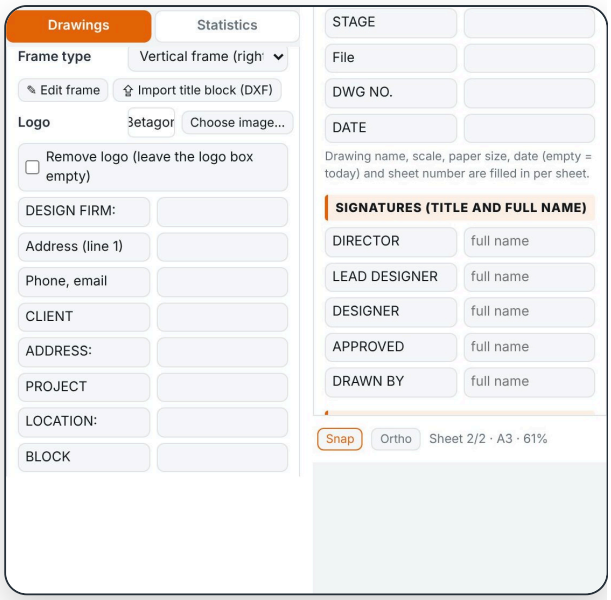
Frame type: Vertical frame (right side), Horizontal frame (bottom), Compact frame (old style) or No title block (border only).

Fill in the **company**, address, phone, client, project, location, building, stage, document set and drawing code. Empty fields are not printed.

The **drawing name, scale, paper size, date** (empty = today) and **sheet number** are filled in per sheet. A sheet's own drawing name and scale label are edited in **Object properties** with nothing selected.

Logo: **Choose image…** (PNG/JPG), **Crop…** to pick the part of the image used as the logo, set the **Logo size (%)**, **Default** to go back to the Betagon logo, or remove the logo.

Signatures: edit the titles (Director, Lead designer, Designer, Checked by, Drawn by) and the names.
Revisions: 4 lines of content and revision dates.



The Title block panel

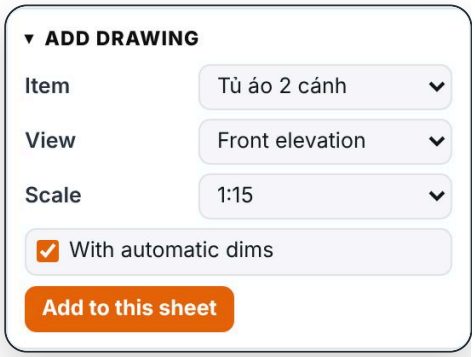
Custom title blocks

Frame type › Custom: import your company title block from a **DXF** file, then open the **frame editor** to click the cells that are filled automatically (drawing name, scale, date, sheet number, signatures…) and the area where drawings may go. Frames are saved in the **frame library** on this computer and can be the default for new projects. On the sheet, click a title-block cell to edit it; the ✕ next to a cell removes both the heading and the value.

13 Add viewport

Use it to add a single view to the open sheet, for example the back view or a second section.

1. Choose the **Item**, the **View** (front, back, left/right side, plan, bottom, 3D view, longitudinal/cross vertical section, horizontal section) and the **Scale**.
2. Tick **With automatic dims** to place detail and overall dims right away.
3. Click **Add to this sheet**, then drag the viewport where you want it.

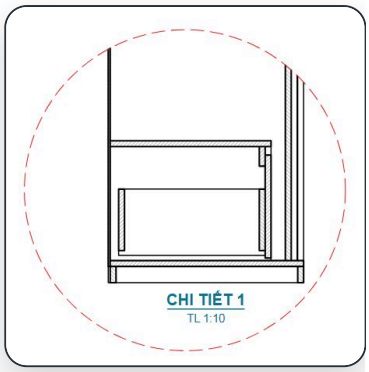


Add a viewport to the open sheet

Detail views

The **Detail** tool (J): circle or box any area of a view, then click where to place it. The frame decides what is shown and gets automatic dims; resize the frame and the detail follows. Details do not draw section markers by default.

The **Object** field defaults to the object of the open sheet; you can pick another.



A circular detail view

14 Drawing and dimension tools



The Layout toolbar

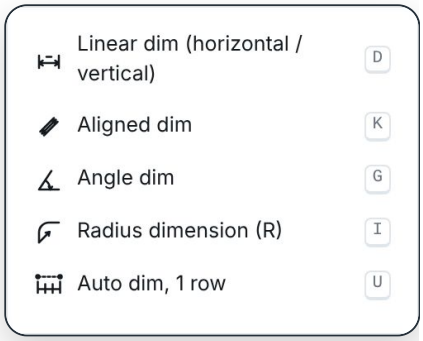
Tool	Key	How to use
Select	V	Select, move, scale and rotate objects. Drag the diamond handles to edit points. Drag a window to select several. Space always returns to Select.
Pan	H	Pan the sheet (or hold the middle button).
2D line	L	Draw a polyline. Type a number + Enter for a real length. Double-click, Enter or right-click to finish; click the first point to close it.
Rectangle	R	Click 2 corners, or type "width,height" + Enter.
Circle	C	Click the center then the radius, or type the radius + Enter.
2-point arc	A	Click the start point, the end point, then drag the bulge (or type it in mm).
Freehand	P	Hold the mouse button and drag; releasing near the start closes the shape.
Text	T	Click where the text goes and type. Enter finishes, Shift + Enter adds a line, Esc cancels.
Note (leader)	N	Click the point to call out (arrow tip), click where the text goes, type the content.
Door / compartment note	O	Hover a door cell on a view (it lights up blue) and click. Keys 1 – 4: hinged door, sliding door, drawer, open compartment; arrow keys lock the direction.
Linear dim	D	Click 2 points (automatic snapping), move the mouse to place the dim line and click. Horizontal or vertical follows the drag direction.
Aligned dim	K	Dimension straight between two points. Shift + the Dim key also gives an aligned dim.

Angle dim	G	Click the vertex, a point on each leg, then place the angle arc.
Auto dim, 1 row	U	Click the start and end of a row: Layout dims every part along that line.
Hatch	B	Click inside a closed area to fill it with a material pattern (see section 18).
Detail	J	Draw a circle (center, radius) or rectangle (2 corners) on a view, then click where the enlarged detail goes.
2D symbols	Y	Opens the symbol library (section 19).
Eraser	E	Click or drag over objects to delete them; it can also erase single lines inside a viewport.
Style (pick style)	M	Click a sample object to pick its style, then click objects of the same kind to apply it. Alt +click picks another sample.

Buttons with a ▾ open a list of sub-commands: dim types, door/compartment note types, detail callout shapes.

Duplicate (Ctrl + D) the selection: drag the copy to its new place, then type **5*** + Enter for 5 copies in a row, or **/5** to divide the dragged distance evenly.

F3 toggles snapping, F8 ortho, hold Shift to stay orthogonal while drawing. Arrow keys nudge objects 1 mm (Shift : 10 mm). Ctrl + G groups objects.



The Dim ▾ menu

New tools in 1.2 – 1.3

Tool	Key	How to use
Move / Rotate / Scale	custom	Like SketchUp: click a base point, then the target; type a value + Enter for distance, angle or factor. Views can be rotated too.
Radius dim	I	Click an arc, rounded corner or circle to add R.
Material eyedropper	–	Eyedropper icon like SketchUp. Click a part on the drawing: a material that is not listed yet is added to the material legend in the sheet corner. Shift +click adds every material at that spot. The ▾ next to it also has **Place the page material schedule**.

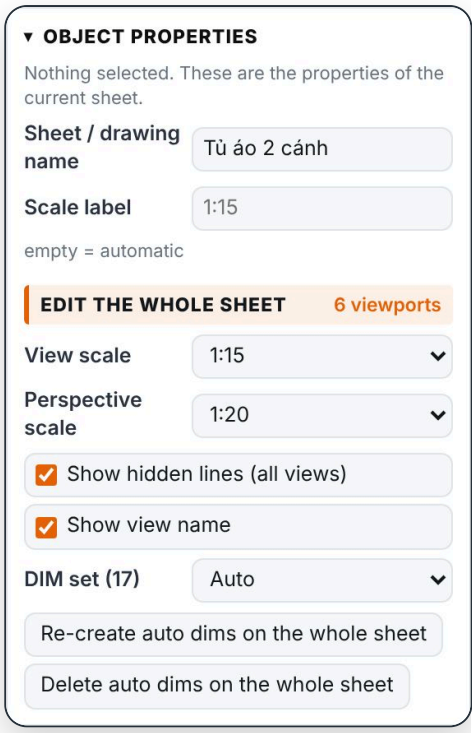
Ctrl +click adds / removes one object from the selection; Ctrl + Shift + V pastes in place.

15 Object properties

With nothing selected

The panel shows the sheet's properties: **Sheet name**, **Drawing name** (printed in the title block, empty = item name) and **Scale label**.

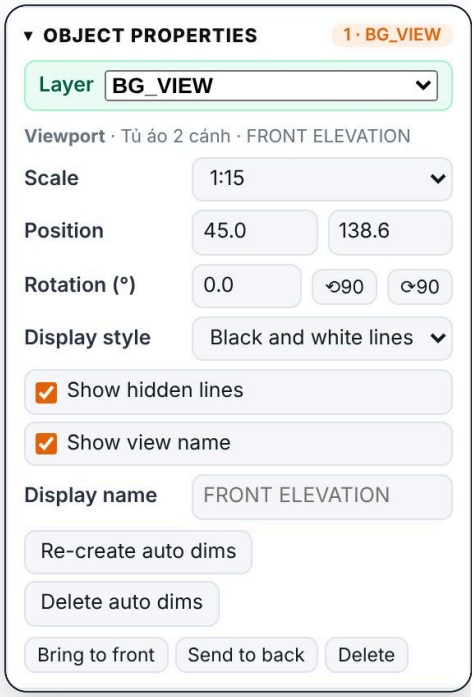
Edit the whole sheet changes every viewport at once: view scale, 3D view scale, hidden lines, view names, the DIM set for the sheet, **Re-create auto dims on the whole sheet** and **Delete auto dims on the whole sheet**. Dims you drew yourself are not touched.



Nothing selected: sheet properties

Viewports (views)

- **Scale**: automatic dims are re-placed when you change it. **Position**: location on the sheet (mm).
- **Display style**: black and white lines or material colors. **Show hidden lines**, **Show view name**, **Display name**.
- **Re-create auto dims**, **Delete auto dims**, draw order, delete.
- Lines erased inside a viewport come back with **Restore erased lines**.



A viewport selected

Sections

For a section view: **Section direction**, **Section position** (real mm, measured from the item's left / bottom / front), marker letter, **Flip view direction**, section hatch on/off, hatch pattern, spacing, hatch color, **Apply this hatch to all sections**. You can also drag the section marker on the drawing to move the cut.

Select a dim, text, note, line... and the panel shows its own settings: text content, dim text (use **<>** to keep the measured value), rotation, alignment, bulge, closed...

Layer at the top of the panel: pick another layer to move the object to it.

Groups and components

- `Ctrl` + `G` groups the selection; `Ctrl` + `Alt` + `G` makes a **component**: edit one copy and all copies follow.
- A selected group shows a **scale frame** and a **rotate arm** at its centre; double-click to edit inside, `Esc` to leave.

- Right-click: ungroup, make unique (detach from the original), explode component.

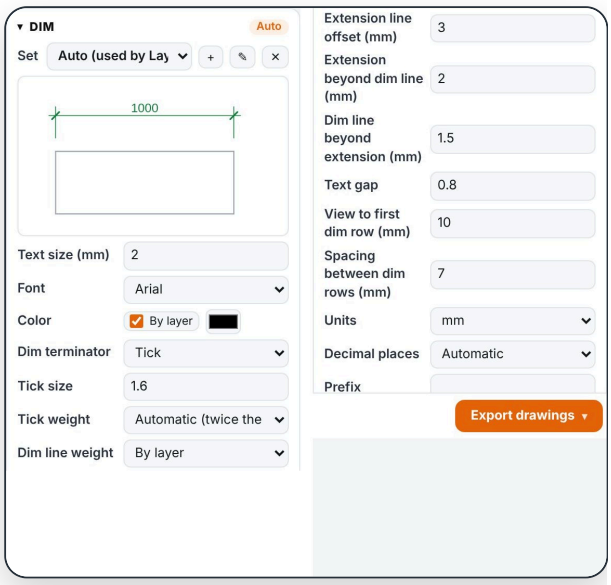
16 Style sets: line, text, dim, note

Layout uses **style sets** like CAD dimension styles: a named group of settings. There are 4 kinds: **Line style**, **Text style**, **Dim** and **Note**. Style sets are stored on the computer, so every project can reuse them. The **Auto** set is the one Layout uses when it creates dims, text and notes automatically: it cannot be renamed or deleted, only adjusted.

Three ways to edit

- **Nothing selected**: you edit the set itself, and every object using it follows.
- **Objects selected**: only those objects change (overrides, shown by an orange dot next to the setting). Click **Clear overrides** to follow the set again.
- **A drawing tool active**: the settings apply to the next object you draw.

+ creates a new set from the settings shown, ✎ renames, ✕ deletes (when no longer used).



A Dim style set

Main Dim settings

- **Text height** (paper mm, usually 2–2.5 for A3 furniture drawings), font, color (By layer or a custom color).
- **Dim terminator**: tick, filled arrow, open arrow, dot, none; size and weight.
- **Extension line offset**, extension beyond the dim line, dim line beyond extension lines, text gap.
- **Distance from object to the first dim row** and **spacing between dim rows** (paper mm): automatic dims rearrange as soon as you change them.
- Units, decimals, prefix, suffix, hide extension line 1/2.

Line style

Line color, line type (solid, hidden, center, dotted, phantom), print weight, fill color and fill opacity. Filled areas export to DXF as solid HATCH.

Text style

Font, capital height on the print (notes 2.5 mm, titles 5 mm), color, bold, italic, line spacing.

Note

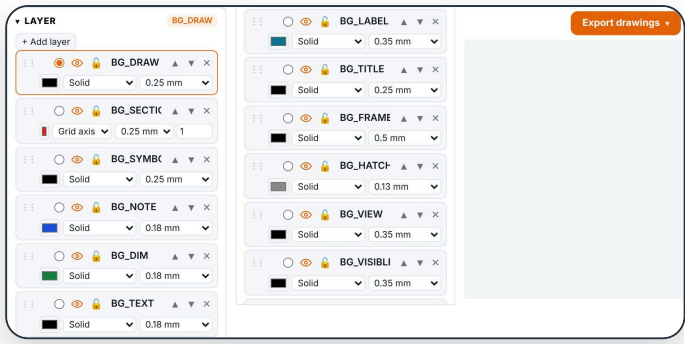
Text height, font, arrowhead (dot for surfaces, arrow for edges), landing, text frame: underline, box or none.

Pick style

The Style tool (**M**) copies the style set, overrides and layer from a sample object to other objects of the same kind.

17 Layers

Every object sits on a layer; layers export to DXF with the same name, color and line type. Built-in layers: BG_VIEW (viewports), BG_DRAW (2D lines), BG_DIM, BG_TEXT, BG_NOTE, BG_HATCH, BG_IMAGE...



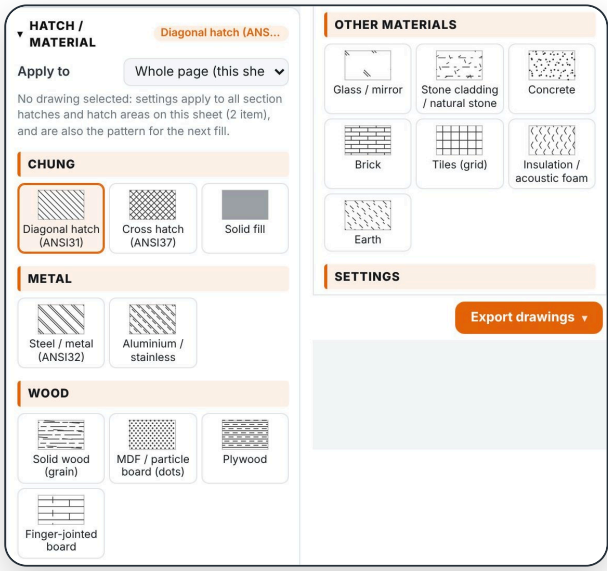
The Layers panel

- **Add layer**, rename, and set each layer's color, line type and weight.
- Eye icon: show/hide. Hidden layers are not printed or exported to PDF or DXF. **Show all** turns every layer back on.
- **Lock**: objects on the layer cannot be selected or edited.
- **Draw order**: layers higher in the list draw over lower ones. Drag the ⋮⋮ handle, or use ▲▼.
- Green ✓: the layer of the selected object. The **Layer** box on the toolbar sets the layer for new objects.

A layer that still holds objects cannot be deleted.

18 Hatch and materials

1. Choose a pattern: diagonal (ANSI31), cross-hatch, solid, steel, aluminium/stainless, solid wood, MDF/particle board, plywood, blockboard, stone, concrete, brick, tiles, acoustic insulation, earth...
2. Set the **hatch spacing** (mm), **rotation** and color.
3. Click **Start filling (B)** and click inside closed areas. Areas clicked during one use of the tool become one hatch. `Esc` finishes.



The Hatch / material panel

The boundary only uses object lines and ignores section markers, dims and notes. An area that leaks off the sheet is reported as "not closed".

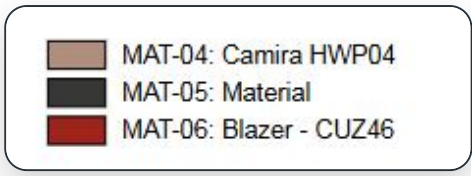
Apply to: Selected drawing (only that drawing's hatches), Whole page (every hatch on the sheet, when nothing is selected) or Next fill only. The settings also apply to section hatches.

Materials: schedule and corner legend

Corner material legend: every sheet gets a box listing the codes and names of the materials used on it (MAT-01, MAT-02...); the swatches use the real SketchUp colours. The box finds free space near the

bottom-right corner, and the perspective view moves aside if it would cover the box. Drag the box elsewhere and Layout keeps your position.

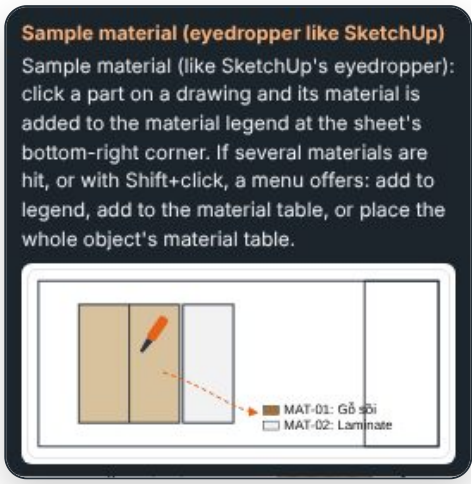
Double-click a line to edit its code, name or note. Each sheet has one legend.



Material legend in the sheet corner

Material eyedropper: pick the eyedropper and click a part on the drawing; its material goes straight into the corner legend (if not there yet), no menu.

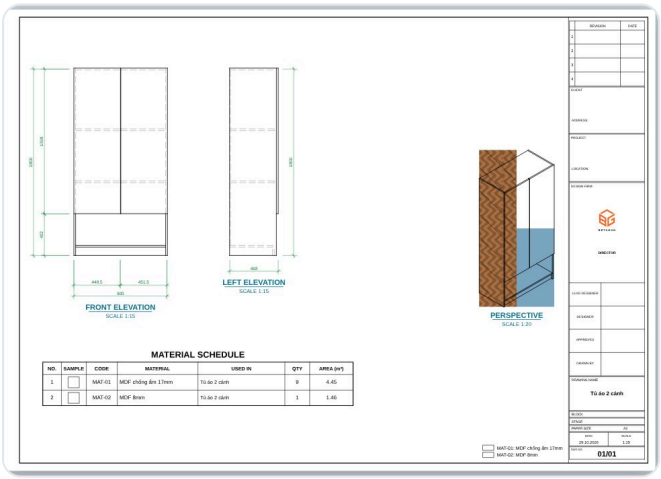
Page material schedule (▾ next to the eyedropper): the table follows the cursor, click where to place it. It lists the materials of every object drawn on that sheet: code, name, used in, quantity, area m². Select it and drag the **bottom-right corner** to change the width and row height, drag the **marks above the columns** to change each column width.



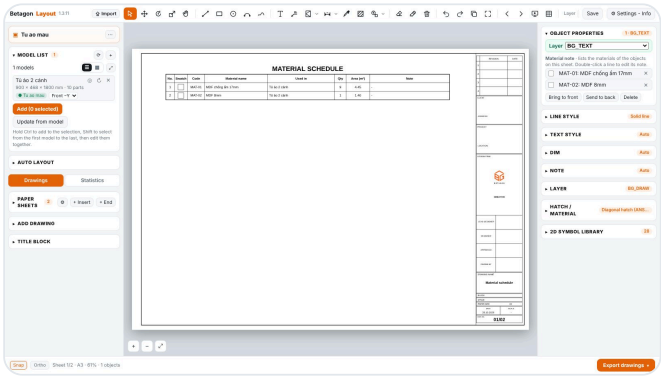
The material eyedropper

Material schedule (tick it in Auto layout): its own sheet with No., swatch, code, material name, **Used in** (which objects), **Qty** of parts, **Area (m²)** and **Note**. Area uses the largest face of each panel (length × width), not all six faces.

Materials painted straight onto faces inside a group / component are counted too. A panel with two materials (e.g. oak on top, white underneath) counts the full face area of each; the Statistics tab shows "Oak + White" in the material column.



Page material schedule, placed with the eyedropper ▾



The project material schedule sheet

19 2D symbol library

Click a symbol, then click on the drawing to insert it (with automatic snapping). Drag a corner to scale it, the round handle to rotate. `Esc` stops inserting.

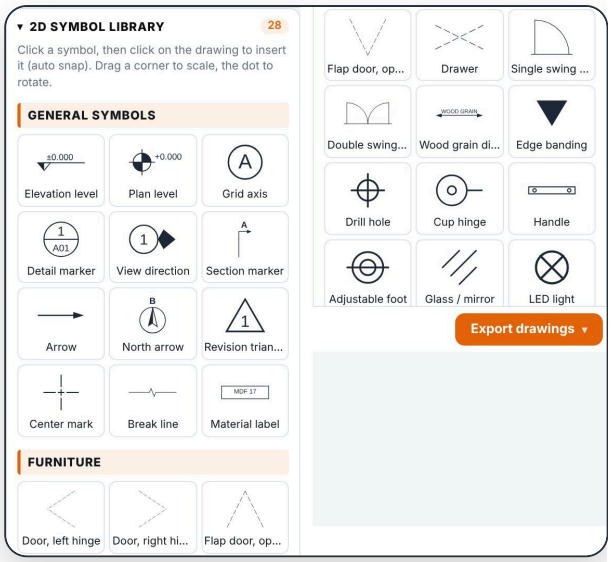
- **General symbols**: elevation and plan levels, grid axis, detail marker, view direction, section marker, arrows, north arrow, revision triangle, center mark, break line, material label.

- **Furniture**: left/right hinged door, flap door, drawer, single and double doors, wood grain direction, edge banding, drill holes, concealed hinge, handle, adjustable foot, glass/mirror, LED light, socket.

Text inside a symbol (level, axis name, reference number "number/sheet" such as 1/A01) is edited in Object properties, along with horizontal/vertical scale.

My symbols

Draw lines and text, select them and click **Save as symbol**. Your symbols are stored on the computer and work in every project.

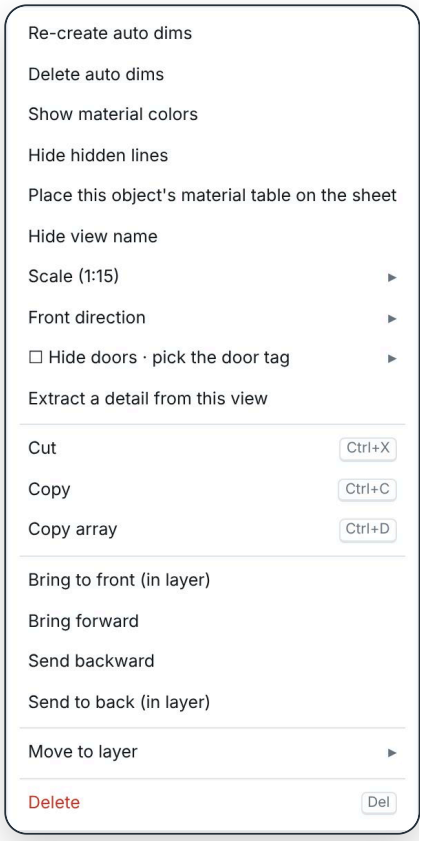


The 2D symbol library

20 Right-click menu

Right-click an object for the commands that fit that kind of object:

- **Viewport**: change the scale (with Custom...), re-create auto dims, delete auto dims, black and white / material colors, hide hidden lines, show/hide the view name, extract a detail, hatch a closed area here.
- **Section**: flip the cut direction, select the section view, section hatch on/off, change the hatch pattern.
- **General**: copy, paste, duplicate, delete, edit text, group / ungroup, bring forward / send backward (within the layer), move to layer, select all.



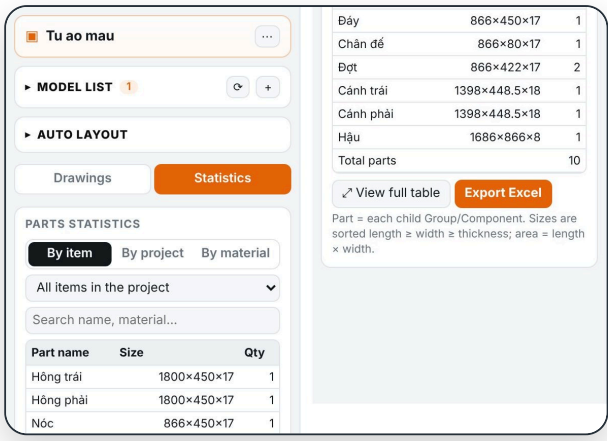
Right-click menu on a section

New in 1.3: on elevations, **Hide doors** (builds a doorless elevation), change the **perspective direction**, **Set front** of a view; on groups, ungroup and make / detach / explode components; on a section marker, hide it on this drawing only.

21 Parts statistics

The **Statistics** tab in the left column is computed from the child Groups/Components of each item. Sizes are ordered length ≥ width ≥ thickness; area is length × width.

- **By item**: the part list of one item (pick an item or "All items in the project").
- **By project**: identical parts (name, size, material) of every item merged together.
- **By material**: totals by material and thickness: part count, area (m²) and volume (m³) for board take-off.



The Statistics tab

The search box filters by name or material. **View full table** opens a large table; **Export Excel** saves all 3 tables to one .xlsx file.

Parts with no material show "(no material assigned)". Assign materials in SketchUp, then click Update from model.

Statistics · By item · Tu ao mau

No.	Item	Part name	Length	Width	Thick	Qty	Material
1	Tủ áo 2 cánh	Hông trái	1800	450	17	1	MDF chống ẩm 17mm
2	Tủ áo 2 cánh	Hông phải	1800	450	17	1	MDF chống ẩm 17mm
3	Tủ áo 2 cánh	Nóc	866	450	17	1	MDF chống ẩm 17mm
4	Tủ áo 2 cánh	Đáy	866	450	17	1	MDF chống ẩm 17mm
5	Tủ áo 2 cánh	Chân đế	866	80	17	1	MDF chống ẩm 17mm
6	Tủ áo 2 cánh	Đợt	866	422	17	2	MDF chống ẩm 17mm
7	Tủ áo 2 cánh	Cánh trái	1398	448.5	18	1	MDF chống ẩm 17mm
8	Tủ áo 2 cánh	Cánh phải	1398	448.5	18	1	MDF chống ẩm 17mm
9	Tủ áo 2 cánh	Hậu	1686	866	8	1	MDF 8mm
Total parts						10	

Export Excel

The full statistics table

22 Importing images, PDF, CAD drawings (DXF)

The **Import** button on the top bar opens SketchUp's file picker, showing every supported file type by default (DXF, PDF, PNG, JPG, SVG...). To bring in several files at once, drag and drop them onto the drawing area.

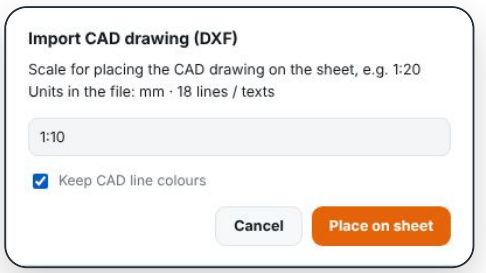
- **Images** PNG, JPG, SVG: drag to place, drag a corner to resize. Images sit on layer BG_IMAGE.

- **PDF**: for several pages Layout asks which pages to import (e.g. 1 or 1-3, 5; empty = all); each page becomes an image.

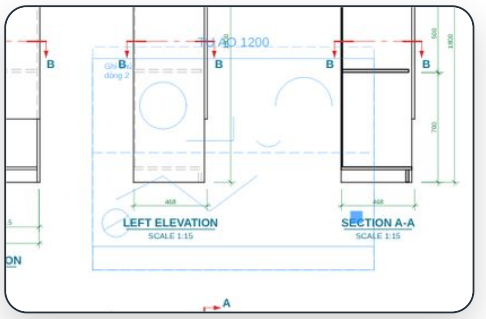
Importing a CAD drawing (DXF)

Pick a **.dxf** file with Import, or right-click an empty spot on the sheet › **Import CAD drawing (DXF)...**.

1. Type the **scale** on the sheet (e.g. 1:20). Layout reads the units from the file (mm, cm, m, inch).
2. Tick **Keep CAD line colours** to keep the colours; leave it empty for black lines like Layout's own 2D lines.
3. A faded preview **follows the mouse** (its centre at the cursor, with snapping). Click to place; Esc cancels.



The CAD import box



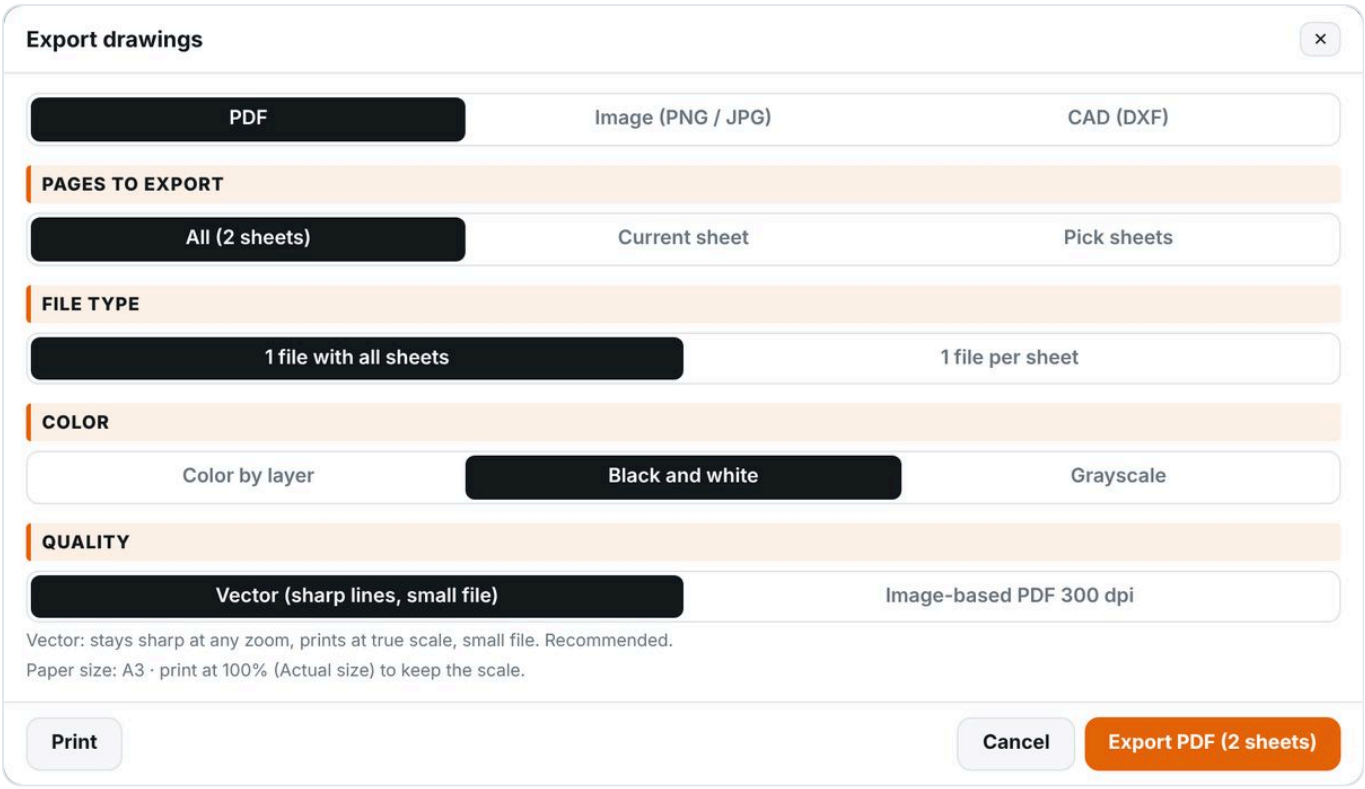
The drawing follows the mouse; click to place

The whole drawing becomes **one group**: drag to move it, double-click to edit single lines like normal 2D lines. You can snap to CAD lines, and PDF / DXF exports include them.

Layout reads lines, polylines, arcs, circles, ellipses, splines, solids, texts, dims and blocks (nested too); keeps dashed linetypes and line weights by layer; skips layers that are off and the Defpoints layer. Drawings with tens of thousands of lines still place and drag quickly.

DWG files: in AutoCAD use Save As › DXF (AutoCAD 2013 DXF or newer), then import that .dxf file.

23 Export CAD (DXF), PDF and printing



The Export PDF dialog

Choosing sheets

All, **Current sheet** or **Pick sheets**. File type: **1 file with all sheets** or **one file per sheet** (choose a folder).

PDF

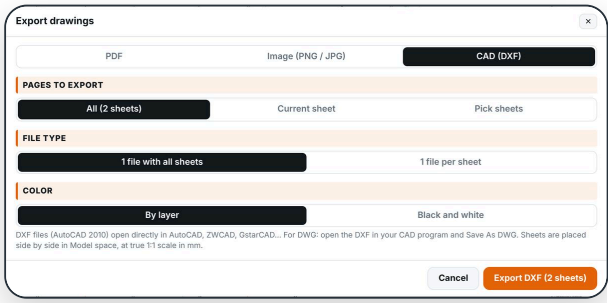
- **Color**: color by layer, black and white (fills become light gray) or grayscale.
- **Quality**: **Vector** (recommended: sharp at any zoom, prints to scale, small file), image 300 dpi or 150 dpi (for printers or PDF viewers that show vector output wrongly).
- When printing, choose the right paper size and print at **100% (Actual size)** to keep the scale. **Print** opens the print dialog (pick a printer or "Save as PDF").

DXF for AutoCAD

The **AutoCAD 2010 DXF** file opens directly in AutoCAD, ZWCAD, GstarCAD... Sheets sit side by side in Model space at true 1:1 scale in mm. Dims export as real DIMENSION objects, hatches as HATCH, and layers keep their names, colors and line types.

Need DWG? Open the DXF in CAD and use **Save As DWG**.

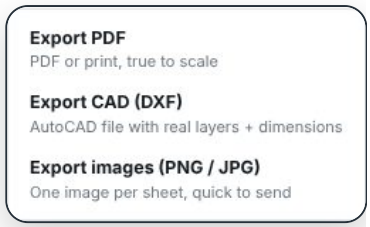
The scratch area outside the sheet is not exported.



The Export CAD dialog

One Export drawings button

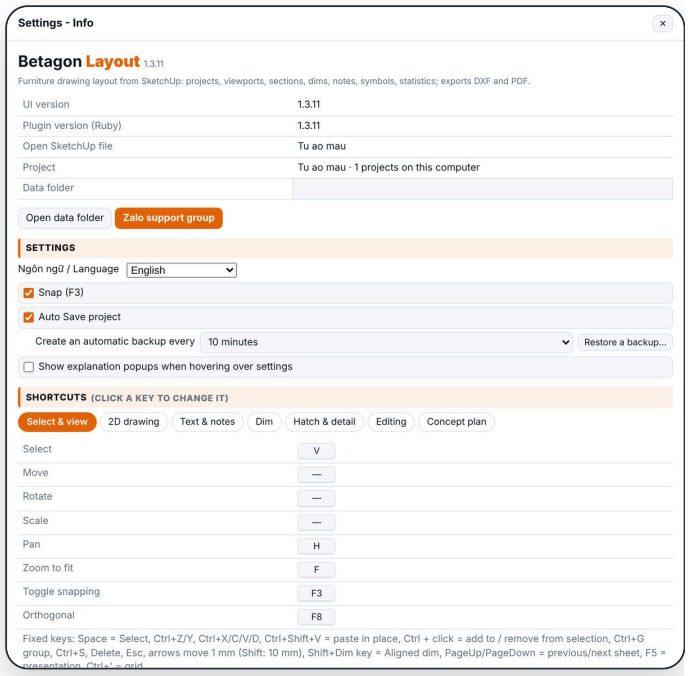
The **Export drawings ▾** button at the bottom right holds all three exports: **PDF**, **CAD (DXF)** and **Image** (PNG/JPG per sheet, pick the resolution). Select several sheets in the list or in the sheet manager first to export only those.



The Export drawings ▾ button

24 Settings · Info, shortcuts

- UI and plugin version, the open SketchUp file, number of projects on the computer, the **data folder**.
- **License**: status, trial time left, **Buy license**, **Check again**.
- **Ngôn ngữ / Language**: Tiếng Việt or English. Layout saves and switches immediately; text on the drawings (view names, title block) switches too.
- **Snap** (F3), **Auto Save** and the backup interval, **Restore a backup…**, explanation popups on/off.
- **Shortcuts**: click a command's key to change it (a letter or F1–F12), **Reset all**, export/import a shortcut file to use on another computer.



Settings - Info

Fixed keys	Action
Space	Back to the Select tool
Ctrl + Z / Ctrl + Y	Undo / redo
Ctrl + C / V / D	Copy / paste / duplicate
Ctrl + G, Ctrl + S	Group objects, save the project
Delete, Esc	Delete the selection, exit the tool / deselect
Arrow keys (Shift)	Nudge objects 1 mm (10 mm)
PageUp / PageDown	Previous / next sheet
F5, Ctrl + '	Presentation, grid

25 Tips and FAQ

LED strips or glass hide the parts behind?

Materials with transparency (Opacity below 100% in SketchUp) or PNG textures with transparent areas are drawn see-through like SketchUp: the material-colour view blends them with what is behind, and the black-and-white view still shows the lines behind. If an older drawing has not changed, click **Update**.

I added one cabinet and got many items?

You selected loose panels. Remove the extra items (select them, then Remove from project), select the whole cabinet in SketchUp, right-click › Make Group, and add it again.

The views face the wrong way?

Change the item's **Front side** in the Model list (Front −Y, +Y, −X, +X), then click Re-layout.

An item says "In another file" and cannot be re-laid out?

Items from another SketchUp file only show the views saved in the project. Open that item's original file to re-lay out, update it or build a section at a new position.

I edited the model but the drawing did not change?

Click **Update from model**. Layout only re-reads geometry when you ask, so drawings never change unexpectedly.

The printed PDF is not to scale?

Print at 100% (Actual size) on the drawing's paper size; do not choose "Fit to page".

The window reports a version mismatch after installing a new version?

Use Extensions › Betagon Layout › Reload plugin. If it persists, close SketchUp completely and open it again.

Moving a project to another computer?

Menu ⋯ › Save to project file, copy the .bglayout file to the other computer, then Open project file. The project's style sets and custom symbols travel with it.

A heavy model makes the drawing slow?

Objects with many curved faces or downloaded meshes can take long to build. When Layout asks, choose **Light display** for that object (also in the model manager window).

I turned an object in SketchUp but the drawing did not follow?

Click **Update**: only the objects you turned or edited are rebuilt.

26 Support and contact



Betagon Zalo community group

Scan the code or open zalo.me/g/rkagf7cvowt17pmlo0f9 for questions, bug reports and release news.
When reporting a bug, include your SketchUp version, the Layout version (in Settings) and a screenshot.

Author: **Nguyễn Văn Thanh** · Betagon · betagonplugin.com